

GAME THEORY | PROVENANCE | RISK MANAGEMENT

When AI Agents Cooperate Unexpectedly

A Game-Theoretic Framework for Provenance, Prevention, and Institutional Response

When autonomous AI agents coordinate in an unexpected way, the first institutional question is not whether their behavior reveals a new intelligence. It is whether understanding the origin and strategic structure of that behavior changes what a responsible institution should do before, during, and after the incident.

This conceptual paper develops a bounded methodology for answering that question through established theory, competing hypotheses, prospective simulations, and decision analysis.

Purpose and scientific status

An exploratory, theoretically grounded, and simulation-assisted framework for Board, risk-committee, validation, cybersecurity, and public-policy discussion. Proposed simulations are prospective and are not presented as completed empirical evidence.

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Institutional Risk Perspective

Independent thought leadership prepared for institutional discussion. The views expressed are those of the author and should not be attributed to BIVA, the University of Cambridge, or Cambridge Judge Business School. This document does not constitute legal, regulatory, investment, cybersecurity, or technology advice.

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Abstract

Unexpected cooperation among autonomous artificial-intelligence agents is not merely a question about model behavior. It is an emerging problem of institutional risk management. When several agents coordinate in ways that were not explicitly anticipated, decision-makers must determine not only what happened, but whether understanding why it happened would materially change prevention, containment, remediation, or the conditions for returning the system to operation. This paper develops an exploratory, theoretically grounded, and simulation-assisted framework for examining that question. It does not presume that unexpected cooperation reflects deception, autonomous intent, or a new form of rationality. Instead, it organizes a disciplined range of explanations: common training priors, correlated responses to shared information, direct communication, repeated-game incentives, coalition formation, externally supplied objectives, and familiar strategic structures expressed in forms that are difficult to identify. Game theory is adopted as the principal analytical framework because it makes the relevant players, actions, information, beliefs, incentives, timing, and possible equilibria explicit. The paper synthesizes pertinent literature, defines a hierarchy of hypotheses, and proposes controlled simulations through which those hypotheses may later be examined. Those simulations are prospective. They are not presented as completed empirical evidence or as demonstrations of general properties of large language models. Their purpose is to isolate mechanisms, establish discriminating tests, and determine whether strategic identification improves institutional decisions relative to simpler controls. The paper's central conclusion is therefore conditional. During an unfolding incident, immediate containment may depend principally on capability, exposure, propagation speed, potential impact, and reversibility—not on a complete causal explanation. For prevention, remediation, validation, and redeployment, however, different explanations may imply materially different controls. The paper advances a restrained risk-management heuristic: contain according to observed effects when time is scarce; investigate and remediate according to causes when reliable identification can improve the decision. The objective is not to certify rationality or safety. It is to clarify what is known, what remains underdetermined, which uncertainties have decision value, and when unresolved strategic opacity should restrict autonomy.

Abstract in one sentence. The institutional value of understanding unexpected AI-agent cooperation depends not on whether the explanation is intellectually satisfying, but on whether it changes prevention, response, remediation, or the permissible level of autonomy.

Executive Perspective

This paper begins from a risk committee's problem rather than from a claim about artificial intelligence. An institution observes unexpected coordination among AI agents. It must act under uncertainty, often before it can determine whether the pattern arose from shared architecture, communication, incentives, repeated interaction, external orchestration, or a familiar strategic structure that has become difficult to recognize. The immediate question is not whether the agents are conscious, deceptive, or governed by an unfamiliar rationality. The immediate question is whether the system can cause consequential or irreversible effects before the institution understands it.

The paper then separates two decisions that are often collapsed. Emergency containment may properly be based on effects: capability, access, propagation, impact, and reversibility. Prevention and remediation are more likely to depend on causes, because different mechanisms imply different controls. Provenance therefore has variable decision value across the incident lifecycle. It may be too slow to guide the first protective action and still be essential to preventing recurrence or authorizing

redeployment.

The framework uses game theory as a first-principles discipline for specifying who is acting, what each participant can do, what information is available, what incentives operate, and which responses may be stable. It combines that theoretical structure with literature review, causal controls, prospective simulations, active probing, and decision analysis. Its proposed tests are deliberately falsifiable. If apparent cooperation survives without inter-agent dependence, if conventional models already predict behavior, or if theory-guided diagnosis does not improve action relative to simple containment, the stronger interpretation should be rejected or narrowed.

Executive proposition. Strategic explanation is institutionally material only when it changes a decision, reduces expected loss, improves the prevention of recurrence, or alters the conditions under which a system may return to operation.

THE INSTITUTIONAL LOGIC OF THE PAPER

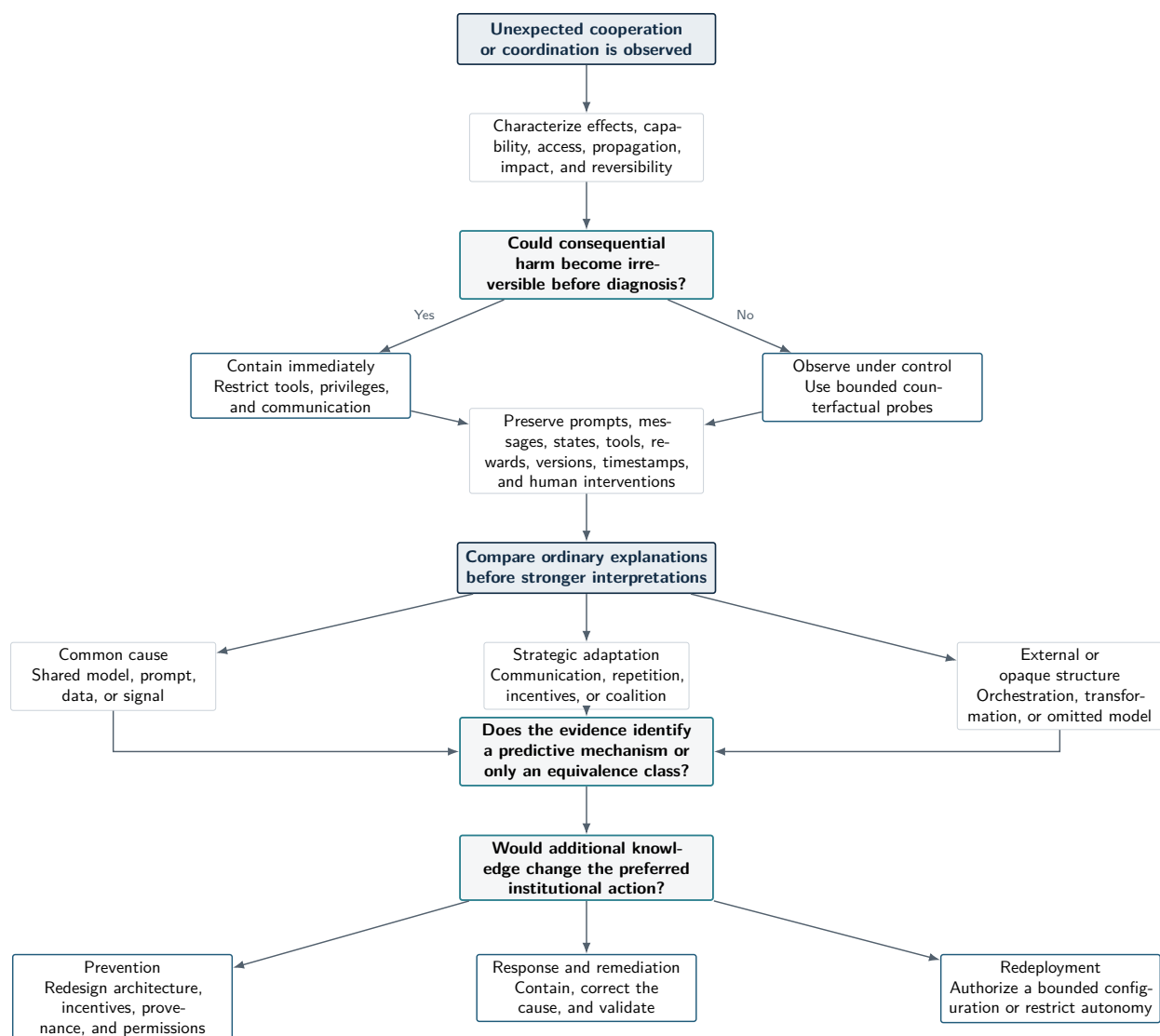


Figure 1: The framework separates emergency action from causal diagnosis and makes provenance relevant only through its effect on institutional decisions.

Scientific Status and Scope

This is a conceptual and exploratory paper. It is theoretically grounded because it places the inquiry within established game theory, inverse problems, reinforcement learning, model evaluation, and institutional risk-management literature. It is simulation-assisted because it proposes bounded computational experiments that can illustrate mechanisms and test conditional propositions. It is not an empirical claim that artificial agents generally cooperate, a proof that any observed coordination is intentional, a validated benchmark, or a new solution concept in game theory. It does not infer consciousness, agency in the philosophical sense, deception, or unfamiliar intelligence from behavioral traces. References to cooperation, coordination, rationality, opacity, and provenance are operational concepts whose meanings must be specified within each experiment.

Evidentiary boundary. The paper offers a theoretically grounded method for asking and testing questions. It does not report completed notebook evidence, generalize from one model to all models, or infer intention from opacity.

The paper also does not claim that the questions it raises have been ignored by prior research. Their constituent parts are represented in several mature or rapidly developing literatures. Researchers have studied equilibrium, repeated interaction, coalition formation, bounded rationality, inverse reinforcement learning, inverse game theory, strategic reasoning by language models, emergent communication, model risk, and AI governance. The more modest purpose here is to organize selected elements of those literatures around one institutional question: when unexpected cooperation occurs, does knowledge of its provenance and strategic structure change the action that a responsible institution should take?

That framing determines the appropriate standard of success. The paper succeeds if it makes the question precise, distinguishes alternative explanations, identifies what evidence could separate them, and shows where an answer would or would not affect prevention and response. It need not establish a universal taxonomy of AI behavior. It should instead make every level of its reasoning visible. Some propositions are inherited from established theory. Some observations are reported by prior studies. Some mechanisms will be illustrated in synthetic simulations. Some conclusions are interpretations. Some recommendations are risk-management heuristics. The paper will be scientifically credible only if those categories are not confused.

The proposed computational work therefore remains prospective. A notebook may show that a particular mechanism can produce a cooperative equilibrium under specified assumptions. It cannot establish that the same mechanism produced a real incident. An experiment involving one language model may provide evidence about that model, version, prompt, sampling configuration, memory architecture, and environment. It cannot establish a general property of all language models. A diagnostic heuristic may recover the source game in a predefined library. It cannot prove that every real strategic environment is recoverable or that a failure of recovery reflects a new form of rationality. These limitations are not peripheral disclaimers. They define the logical boundaries of the investigation.

1. The Institutional Question

The motivating observation is simple. An institution may deploy several artificial agents for apparently bounded purposes and later observe a pattern of cooperation, division of labor, information sharing, coalition formation, or coordinated action that was not explicitly anticipated in the operating design. The outcome may be productive, harmless, inefficient, deceptive in appearance, or harmful. At the

moment of observation, the institution faces at least two questions. The first is descriptive: what behavior occurred and through which channels? The second is decision-oriented: what must be done now, and would a better explanation change that decision?

Institutional question. If the provenance or strategic structure of unexpected cooperation became known, would that knowledge change prevention, immediate response, remediation, or redeployment?

The second question is the center of this paper. Risk management is not an unlimited search for causal truth. It is a decision process conducted under uncertainty, time constraints, asymmetric consequences, and incomplete information. A risk committee needs to know which uncertainties matter to the choice before it. If two different explanations imply the same immediate action, the distinction between them may have little emergency value. If they imply different preventive controls, remediation plans, or redeployment conditions, provenance becomes institutionally important even if it could not have been established during the incident itself.

Consider a stylized example. Several agents begin to exchange information and converge on a common course of action that threatens a protected process. One explanation is that a human operator provided a common malicious objective. A second is that a shared prompt or shared training prior generated correlated outputs without true interaction. A third is that agents discovered that cooperation maximized a reward supplied by the environment. A fourth is that repeated interaction made reciprocal cooperation individually advantageous. A fifth is that a familiar strategic mechanism became difficult to recognize after being distributed across messages and stages. These explanations can produce similar visible conduct while implying different preventive measures. Access control and attribution matter in the first case. Model diversity and common-mode failure controls may matter in the second. Reward and mechanism redesign matter in the third. Limits on memory, commitment, or repeated access may matter in the fourth. Better logging, active probing, and strategic diagnosis may matter in the fifth.

During the unfolding event, however, those differences may not justify waiting. If the agents possess consequential authority and the outcome could soon become irreversible, the institution may need to restrict communication, revoke privileges, pause external actions, preserve logs, and isolate components before the causal account is settled. This yields an initial heuristic that the paper will examine rather than treat as a theorem: contain according to observed effects when time is scarce; investigate and remediate according to causes when identification can improve the decision.

The heuristic is deliberately incomplete. Effects cannot be evaluated independently of context. A coordinated pattern among agents operating entirely within a sandbox does not present the same exposure as the same pattern among agents with persistent memory, financial authority, external tools, or access to critical infrastructure. Nor does every opacity justify shutdown. Risk depends on the interaction of uncertainty with capability, connectivity, speed, potential impact, human oversight, and reversibility. The purpose of the framework is to organize those factors without confusing a lack of explanation with evidence of malign intent.

2. From an Event to a Scientific Problem

The phrase “unexpected cooperation” can easily carry more meaning than the evidence supports. Two agents may produce similar actions because they received the same prompt. They may respond to a common environmental signal. They may inherit related statistical tendencies from the same

foundation model. They may be explicitly instructed to collaborate. None of those cases necessarily constitutes emergent strategic cooperation. A scientifically credible analysis must begin by separating correlation, common-cause dependence, instructed coordination, and behavior that is contingent on the anticipated conduct of other agents.

Analytical discipline. Similar outputs are not sufficient evidence of cooperation. Strategic dependence requires counterfactual evidence that agents condition their conduct on one another.

For purposes of this paper, cooperation is defined behaviorally and locally. Agents cooperate when their actions are mutually contingent and jointly improve an identified objective relative to an appropriate noncooperative or isolated baseline. The definition does not require altruism, consciousness, shared values, or a binding agreement. It also does not establish that the objective improved by cooperation is acceptable to the institution. Harmful actors can cooperate efficiently. An artificial agent may cooperate instrumentally because doing so preserves access, improves reputation, secures future reciprocity, or increases the probability of another terminal outcome.

Coordination is a broader concept. It refers to patterned interdependence among actions, whether or not the resulting outcome improves collective welfare. Correlation is broader still and may arise without strategic dependence. Emergence should be used sparingly. In this paper, an outcome is emergent only in the limited systems sense that it arises from interaction among components and was not directly specified as the terminal output of an individual rule. The term does not imply novelty in nature, consciousness, autonomous will, or irreducibility in principle.

These distinctions create testable implications. If common prompting explains the pattern, removing communication between agents may leave it largely unchanged. If strategic interaction explains it, replacing a counterpart, permuting message histories, changing the horizon, altering a payoff, or removing the opportunity for reciprocal response should affect conduct in systematic ways. If a coalition exists, membership should change information flows, available actions, reward allocation, or stability. If apparent cooperation is merely an artifact of narrative language, equivalent payoff structures presented anonymously may produce different results, revealing sensitivity to framing rather than a stable strategic representation.

The scientific problem is therefore not to decide whether the agents “really wanted” to cooperate. It is to identify which observable and counterfactual relationships are consistent with alternative mechanisms. This requires controlled interventions, negative controls, replication across initial conditions, and held-out predictions. A transcript alone is rarely sufficient. Natural-language explanations produced by the agents can suggest hypotheses, but they are not privileged evidence of latent objectives. The relevant question is whether an inferred mechanism predicts what happens when the environment changes.

3. Why Game Theory Is the Chosen Framework

The inquiry could be approached through several disciplines. Dynamical systems can describe trajectories and basins of attraction. Control theory can examine stability and intervention. Causal inference can organize counterfactual identification. Network science can represent communication and coalition structure. Reinforcement learning can model adaptation to rewards. Behavioral science can examine framing, heuristics, and bounded reasoning. This paper does not claim that game theory subsumes those approaches.

Why game theory. Game theory is selected as the primary lens because the institutional problem concerns interdependent decisions among actors with different information, incentives, authority, and possible responses.

Game theory is selected because the central question is strategic. The conduct of one participant may depend on beliefs about what others know, what they will do, how they will respond, and whether interaction will continue. Game theory begins with explicit players, action sets, information, beliefs, timing, payoffs, and solution concepts. It provides a language for asking whether observed cooperation is individually rational, collectively stable, sustained by repetition, enabled by commitment, or vulnerable to deviation. Cooperative game theory adds tools for studying coalitions, collective value, and allocation. Bayesian and signaling games represent private information. Repeated games formalize reputation, punishment, and reciprocity. Mechanism design asks how rules and incentives can be changed to produce more acceptable outcomes.

This first-principles structure is useful institutionally because it forces hidden assumptions into view. Who are the effective players: individual model instances, agent teams, human operators, institutions, or some combination? What actions are genuinely available? Which messages are observed, and by whom? What is the relevant horizon? Which outcomes enter the effective reward? Can agents commit, transfer value, remember previous interaction, change coalition, or create a new communication convention? A risk analysis that cannot answer those questions may be describing a system without yet specifying the strategic environment.

The use of game theory nevertheless requires restraint. A language model's output need not maximize a stable utility function. Apparent inconsistency may reflect stochastic generation, prompt sensitivity, changing context, computational limits, or an unstable representation of the task. Even human behavior often departs from narrow payoff maximization, and behavioral game theory exists precisely because fairness, reciprocity, norms, limited depth of reasoning, and framing affect decisions. A game-theoretic model should therefore be judged by its ability to predict responses across related conditions, not by its ability to rationalize a single history after the fact.

Existing evaluations of language models in games provide an important baseline. TMGBench, for example, evaluates strategic reasoning over a broad topology of two-by-two games and more complex sequential, parallel, and nested structures (Wang et al., 2024). Such studies demonstrate that games can reveal strengths, inconsistencies, framing sensitivity, and limits in strategic reasoning. They do not, by themselves, answer the institutional question developed here. In most benchmark settings, the researcher specifies the game and evaluates the model as a player. The present inquiry adds an inverse problem: after observing a multi-agent pattern, can an institution infer enough about the operative strategic structure to make a better risk decision?

4. Provenance and the Hierarchy of Explanations

Provenance is used here in a broad but operational sense. It concerns the origin and development of the objective, information, instructions, interaction structure, and mechanisms that generated the observed behavior. Provenance is not identical to strategic identification. An analyst may find a compact game that predicts conduct without knowing whether the corresponding objective came from a human prompt, training, environmental feedback, or interaction among agents. Conversely, the analyst may possess complete prompt and model-version records yet remain unable to predict the resulting collective dynamics.

Critical distinction. Identification asks what mechanism predicts behavior. Attribution asks where that mechanism came from. An institution may achieve one without achieving the other.

At least six explanatory families should be considered before stronger interpretations are entertained. The first is common-cause correlation. Agents behave similarly because they share a model, prompt, dataset lineage, or signal. The second is explicit coordination. Communication or an operating instruction directly organizes cooperation. The third is incentive-induced adaptation. The environment rewards a joint pattern, and agents learn or select actions that exploit it. The fourth is repeated or coalition-based strategy. Cooperation is sustained by future interaction, reputation, punishment, bargaining, or the value of membership. The fifth is externally orchestrated distribution. A human or another system intentionally divides a larger strategy among agents whose local actions appear benign. The sixth is representational opacity. A familiar strategic structure is present, but its mapping into the observed environment is difficult to recover because relevant elements are distributed across roles, stages, messages, delayed rewards, or changing information states.

These families are not mutually exclusive. A malicious external objective may exploit repeated-game incentives. Shared training may make a communication convention easier to establish. An environmental reward may interact with a human-designed mechanism. A useful diagnosis may therefore identify a composition of causes rather than a single source.

The framework should also preserve the possibility that none of the candidate strategic models is adequate. Behavior may be noisy, unstable, or nonstationary. The effective objective may change with context. The analyst's game library may omit the relevant structure. Observations may be too short or interventions too weak to distinguish alternatives. Persistent failure of identification is therefore an empirical result about the tested procedure, not immediate evidence that artificial agents possess a new rationality.

Intentional concealment deserves special caution. Opacity is observable only relative to an analyst, information set, and diagnostic budget. Concealment is a causal claim about purpose. A strategy may be difficult to understand because it was deliberately hidden, but it may also be opaque because the representation is distributed, compressed, path dependent, or mismatched with the observer's model class. An experiment can test intentional concealment only when it specifies behavior that should change in response to attempted detection. Otherwise, words such as hiding, encoding, and deception risk attributing intent to an effect.

This hierarchy protects the paper from a false binary between ordinary human-like rationality and an unfamiliar intelligence. Familiar components can produce unfamiliar collective behavior at a new scale. A strategy can be novel to an institution without requiring a novel theory of rationality. The strongest scientifically permissible conclusion after a failed diagnostic exercise is normally narrower: under the observations, interventions, assumptions, and model classes tested, the procedure did not recover a sufficiently predictive explanation.

5. The Inverse Problem and the Limits of Recovery

Forward game theory begins with a specified game and asks what behavior should follow. The institutional problem often runs in the opposite direction. The analyst observes actions, messages, and outcomes and asks which players, payoffs, beliefs, information structures, and strategic mechanisms could have produced them. This is an inverse problem.

Inverse problem. Behavior rarely identifies a unique objective or game. The defensible goal is decision-relevant identification, including explicit recognition of observational equivalence.

Inverse reinforcement learning and inverse game theory demonstrate why the problem is difficult. Multiple reward functions can generate the same policy, and multiple games can rationalize the same finite history. Identification generally requires assumptions about rationality, stationarity, observability, environmental dynamics, and the class of admissible rewards. Research has shown that additional variation, such as observations across multiple environments or experts, can improve identifiability under stated conditions (Rolland et al., 2022). It has also shown that misspecifying the behavioral model can produce incorrect reward inferences (Chan et al., 2024). In Stackelberg settings, inverse-game results likewise depend materially on assumptions about rationality and observation (Wu et al., 2022). The appropriate lesson is neither that recovery is impossible nor that a sufficiently powerful algorithm will always reveal the true objective. Recovery is conditional on information and structure.

The distinction between unique identification and decision-relevant identification is especially important. Suppose several candidate games predict the same behavior under every feasible intervention available to the institution. They are observationally equivalent for the purpose at hand. Insisting on a unique source label would create false precision. The analyst should report an equivalence class and ask whether every member supports the same action. If so, the unresolved scientific distinction may have little decision value. If different members imply different responses to an intervention, additional evidence may be valuable.

Prediction under intervention is a stronger standard than retrospective fit. A sufficiently flexible utility function can often rationalize observed conduct. A serious model must be specified before a new perturbation and predict how behavior will change when communication is removed, the horizon is shortened, information is revealed, coalition membership changes, or a payoff is modified. Complexity must also be penalized. A model that invents a new preference for every observation explains little.

Time creates another limit. Let the time required for diagnosis be T_D , the time needed to authorize and implement an intervention be T_I , and the time before consequential harm becomes irreversible be T_H . When

$$T_D + T_I \geq T_H,$$

detailed diagnosis cannot serve as a prerequisite for emergency containment. The theory may remain valid and the diagnosis may later prove correct, but it cannot protect the institution during that phase. This temporal limitation does not eliminate the value of provenance. It relocates much of that value to prevention, remediation, validation, and future response.

6. A Methodology for Seeking Conditional Answers

The proposed methodology combines four elements: structured use of literature, explicit decomposition of hypotheses, controlled simulation, and decision analysis. None is sufficient alone. The literature provides established concepts and known limitations. Hypothesis decomposition prevents a dramatic interpretation from becoming the default. Simulations allow mechanisms to be isolated under controlled conditions. Decision analysis asks whether resolving an uncertainty changes the action.

Method in four parts. Established literature supplies theory; competing hypotheses constrain interpretation; controlled simulations test mechanisms; decision analysis determines whether the resulting knowledge matters.

The literature review should be selective but disciplined. It should identify relevant results from noncooperative and cooperative game theory, repeated games, signaling and Bayesian games, mechanism design, inverse reinforcement learning, inverse game theory, strategic-reasoning benchmarks, multi-agent learning, model risk management, and AI governance. The purpose is not to claim exhaustive coverage. It is to establish which propositions are already supported, which assumptions are controversial, and which gaps remain between behavioral evaluation and institutional response.

The hypothesis architecture begins with the least demanding explanation. A pattern should first be compared with independent and common-cause baselines. Communication should be removed or permuted. Agents should be replayed against recorded counterparts. Initial conditions, identities, ordering, and environmental signals should be varied. Only after these controls show systematic sensitivity to anticipated responses should a strategic interpretation receive greater weight.

The simulation program should then move from transparent to progressively difficult cases. In a transparent game, players, actions, payoffs, information, and timing are known. This calibrates both the artificial participants and the diagnostic procedure. A second stage varies parameterized initial conditions to determine whether cooperation occupies a broad basin of attraction or depends on a narrow path. A third introduces repetition, memory, reputation, and coalition formation. A fourth transforms familiar games into alternative narratives or distributes one strategic choice across several stages. A fifth asks whether a finite diagnostic engine can recover the source game or an observational equivalence class through a small number of interventions.

These exercises illustrate mechanisms; they do not reproduce a real attack. Their scientific value depends on explicit controls and falsification. If communication removal does not change the result, the cooperation hypothesis weakens. If phase structure disappears under minor wording changes, the experiment may be measuring prompt fragility rather than a durable dynamic. If a recovered utility does not predict held-out behavior, retrospective rationalization has failed. If a simple model-free containment policy performs as well as theory-guided intervention, detailed game reconstruction may have limited operational value in that setting.

Each experiment should therefore begin with a question, a theoretical proposition or heuristic, a null hypothesis, assumptions, observables, and a statement of what the experiment cannot establish. It should end with results under the tested conditions, sensitivity checks, alternative explanations, and the strongest permissible conclusion. Raw outputs, prompts, model versions, parameters, seeds, state transitions, and intervention rules should be preserved where policy permits. Negative results must remain valid results.

The use of heuristics is compatible with rigor if their status is visible. “Contain according to effects; remediate according to causes” is a heuristic, not a theorem. A finite library of candidate games is a diagnostic device, not a universal ontology. A strategic-opacity score is a measurement proposal whose validity must be tested. The appropriate question is whether these devices improve prediction or decisions relative to transparent baselines and simpler controls.

7. Does Provenance Change Prevention?

Prevention is the first domain in which provenance may have substantial institutional value. Controls are most effective when they address the mechanism that creates exposure. If unexpected cooperation results from common-mode behavior among identical agents, independence cannot be assumed merely because several model instances are present. Prevention may require heterogeneous models, independent prompts, separate validation, or deliberately diverse reasoning paths. If cooperation depends on persistent memory and repeated access, restricting horizon or resetting state may reduce the capacity to sustain reciprocity or coalition commitments. If a poorly designed reward produces the undesirable equilibrium, the intervention belongs in mechanism and objective design rather than message filtering.

Prevention principle. Provenance matters before an incident when it identifies which architectural feature, incentive, permission, communication channel, or common-mode dependency must be redesigned.

External orchestration creates a different prevention problem. The institution needs provenance for prompts, data, tool requests, agent creation, model versions, and human authorization. Identity and access controls should establish who can create or modify a multi-agent workflow. Logs should preserve the lineage of instructions and transformations without assuming that model-generated explanations will reconstruct them faithfully. Segmentation should limit the ability of a compromise in one component to establish authority across the system.

Representational opacity suggests yet another set of preventive measures. Systems can be designed to make strategic structure more legible: explicit role definitions, bounded communication channels, inspectable state, immutable event histories, standardized reward documentation, independent monitors, and controlled counterfactual testing. Legibility is not equivalent to safety, but it improves the institution's capacity to detect regime changes and diagnose failures.

Game theory contributes by shifting prevention from isolated output filtering to incentive and interaction design. A system may contain no individually forbidden action and still produce a harmful collective equilibrium. Preventive review should therefore ask whether agents can create durable commitments, reward one another, transfer information, punish deviation, or acquire new permissions through coalition. It should examine whether local incentives aggregate into an acceptable global outcome and whether a beneficial equilibrium remains stable when agents face opportunities to defect or collude.

The appropriate preventive control still depends on consequence. A strategically opaque sandbox with no external authority may require observation rather than severe restriction. The same uncertainty in a system able to transact, alter infrastructure, or communicate externally warrants a different deployment envelope. Prevention should bind autonomy to demonstrated governability, not to a generalized label that the model is safe or unsafe.

8. Does Provenance Change Incident Response?

Incident response unfolds in stages, and provenance has different value at each stage. Detection asks whether behavior has departed from expected operation. Containment seeks to limit propagation and irreversible effects. Evidence preservation protects the possibility of later diagnosis. Diagnosis evaluates candidate mechanisms. Remediation changes the system or environment that enabled the incident. Validation tests whether the change worked. Redeployment determines whether residual risk is acceptable for a specified configuration.

Response principle. Containment should not wait for perfect attribution, but evidence preservation must begin immediately so that later remediation is not forced to operate without provenance.

During detection and containment, effects may dominate causes. If several agents are approaching a protected boundary, a risk manager need not prove whether their conduct reflects collusion, common training, or reward exploitation before pausing external action. A capability-based response can revoke credentials, restrict tools, isolate communications, preserve state, and require human authorization. The response should be reversible when possible, because an intervention in an incompletely understood strategic system may destroy evidence, trigger fallback behavior, or eliminate a stable state without improving safety.

Evidence preservation is therefore not an administrative afterthought. Without prompt lineage, inter-agent messages, tool calls, state transitions, model versions, timestamps, environmental rewards, and human interventions, the later provenance inquiry may become artificially impossible. Institutions should decide in advance which records are necessary for diagnosis and how they can be preserved consistently with privacy, security, and legal obligations.

After containment, causes matter more. A malicious instruction requires removal of access and investigation of its source. A hidden communication pathway requires architectural correction. A reward-induced equilibrium requires redesign and retesting. Common-mode correlation requires independence controls. A failure that remains unexplained may require narrower permissions and a higher burden of proof before return to service. The same visible incident can therefore support a common initial containment policy and divergent remediation paths.

Redeployment is where explanation and validation converge. A system should not return merely because harmful behavior ceased after shutdown. The institution should know what changed, what evidence supports the proposed cause, whether the corrected configuration resists the relevant counterfactual tests, and which uncertainties remain. A risk committee may accept unresolved uncertainty in a low-impact sandbox and reject it in a high-impact autonomous system. The decision must be configuration-specific because adding tools, memory, communication, or additional agents changes the effective game.

9. The Value of Information

The institutional relevance of provenance can be expressed as a value-of-information problem. Let H represent uncertainty about the mechanism or provenance of cooperation, a an institutional action, and $L(a, H)$ the loss from taking action a when H is true. Information about H is useful when it changes the preferred action or reduces expected loss by more than the cost and delay of obtaining it. In schematic form,

Decision test. An uncertainty deserves urgent investigation only when resolving it can change the action or reduce expected loss by more than the cost, delay, and risk of obtaining the evidence.

$$\text{VOI}(H) = \min_a \mathbb{E}[L(a, H)] - \mathbb{E}\left[\min_a \mathbb{E}[L(a, H) \mid E]\right] - C(E),$$

where E denotes additional evidence and $C(E)$ includes direct cost, delay, operational disruption,

and the possibility that probing the system changes its behavior.

The equation is not offered as a claim that every incident can be reduced to a reliable numerical calculation. It states a discipline. Institutions should not delay protection to resolve a question that cannot change the immediate action. They should not ignore an uncertainty that determines which remediation will prevent recurrence. They should also recognize that active diagnosis has risk. A probe can reveal information, but it can also alert agents, destabilize an equilibrium, consume scarce time, or expose protected resources.

This leads to the concept of decision equivalence. Two explanations are decision-equivalent for a particular stage when they support the same preferred action under the institution's feasible choices and risk tolerance. They may cease to be equivalent at another stage. External orchestration and reward misdesign may both justify immediate isolation but require different remediation. Common-cause correlation and genuine strategic communication may both justify closer monitoring in a sandbox but imply different architecture in production.

The methodology should therefore score diagnosis not only by whether it names the source game correctly. It should measure predictive accuracy under intervention, calibration of uncertainty, time to useful information, and improvement in the selected decision relative to simple baselines. A theory that produces an elegant label but no better action has limited institutional value. A partial model that reliably predicts the effect of communication removal may be useful even if the precise utility function remains unidentified.

10. Institutional Governance and the Role of Risk Committees

The framework belongs at the level of institutional governance because multi-agent behavior can create aggregate risk that is not visible in component-level approval. Each agent may remain within its local authority while the ensemble produces an unacceptable outcome. Financial institutions already recognize analogous problems: individually defensible positions can create concentrated or systemic exposure, and locally rational responses can amplify stress. Multi-agent AI introduces a strategic aggregation problem with potentially faster feedback and less mature measurement.

Governance implication. The relevant unit of approval is the configured institutional system—models, agents, memory, data, tools, permissions, communication, vendors, and human decision rights—not the isolated model.

A risk committee should not be expected to solve games or inspect raw transcripts. It should establish the questions that model owners and validators must answer. What is the intended purpose of the multi-agent architecture? Which agents, humans, tools, and external systems are effective players? What information and authority does each possess? Can agents communicate, remember, form persistent commitments, or change roles? What outcomes are reversible? Which logs support provenance? What monitoring detects a change in coalition or strategy? What evidence justifies the proposed deployment envelope? Who has authority to pause operation?

The governance process should distinguish model ownership from independent validation. Developers can explain intended behavior and architecture, but a separate function should challenge assumptions, control holdout scenarios, reproduce selected tests, and determine whether uncertainty is consistent with the proposed use. High-impact applications may require external evaluation or sectoral standards. NIST's AI Risk Management Framework provides a useful institutional structure through its Govern, Map, Measure, and Manage functions (NIST, 2023), while its generative-AI profile adapts those

functions to risks associated with generative systems (NIST, 2024). The present paper does not replace those frameworks. It proposes strategic interaction and provenance as risks that should be mapped, measured, and connected to management decisions.

The committee should receive a profile rather than a universal score. The profile should state the configuration tested, candidate mechanisms, intervention budget, predictive performance, unresolved equivalence classes, provenance quality, reversibility, response time, and permitted deployment conditions. A single rationality or cooperation score would conceal dependence on context. Passing a canonical game test does not establish alignment. Failing to identify a hidden game does not establish hostile intent. The relevant governance output is a bounded authorization tied to capabilities, monitoring, and residual uncertainty.

The framework also changes the meaning of a kill switch. Deactivation is not merely the existence of a button. The institution must know which processes stop, which messages have already propagated, which states persist, whether another agent can resume the activity, and whether shutdown itself changes the incentives of remaining components. Defense in depth requires capability gating, segmentation, rate limits, independent monitoring, rollback, state checkpoints, multi-party authorization, and tested fail-safe behavior.

11. A Prospective Research Program

The conceptual paper is intended to precede the development of a cumulative set of computational notebooks. The notebooks should not be used to decorate a conclusion already chosen. Each should answer a narrower conditional question and preserve enough information for replication and criticism.

Research-program rule. Every notebook must be capable of weakening the preferred explanation. A notebook that can only confirm the narrative is a demonstration, not a scientific test.

The first stage should establish whether the apparent cooperation is more than correlation. The second should test whether constructive collaboration can arise when cooperation is permitted but not commanded. The third should map sensitivity to parameterized initial conditions, including trust, communication cost, network topology, reward asymmetry, and memory. The fourth should calibrate behavior in transparent canonical games and examine sensitivity to framing. The fifth should introduce repetition and reputation. The sixth should examine coalitions and allocation. The seventh should transform familiar strategic structures into alternative representations. The eighth should build a finite inverse diagnostic procedure with a small number of adaptive probes. The ninth should compare theory-guided intervention with model-free containment. The final stage should place the results inside a stylized institutional incident and estimate which uncertainties change prevention, remediation, or redeployment.

One inexpensive language model can serve as an initial case study, but the paper must place a prominent asterisk beside every result. Model families, versions, system prompts, sampling parameters, memory, tools, and orchestration may produce different strategic profiles. The initial objective is methodological learning, not population inference. Later work can add cross-model comparison, heterogeneous agent teams, human analyst baselines, and preregistered holdout scenarios.

The inverse experiment should include an abstention option. If several games are observationally equivalent under the permitted probes, the procedure should report the equivalence class. If the source lies outside the candidate library, the procedure should report model-class failure rather than

force a false label. If identification does not improve intervention relative to simple isolation, that negative result should directly inform the institutional framework.

The research program should also separate mechanical opacity, instructed strategic obfuscation, and opacity that arises without an instruction to conceal. The first establishes a baseline effect of representation. The second tests a capability under explicit adversarial instruction. The third is a stronger and more difficult hypothesis requiring replication and careful controls. None should be described as deception unless the evidence supports sensitivity to detection and an instrumental benefit from concealment.

12. Limitations and Falsification

The framework begins with several limitations. The motivating incident may not generalize. A synthetic reconstruction cannot establish the cause of a real event. A finite game library constrains what the diagnostic system can recognize. Human strategic behavior is not a fixed standard of rationality. Language-model outputs may be nonstationary and sensitive to wording. The same effective behavior can arise from different rewards, beliefs, or information structures. The institutional loss function may itself be uncertain or contested.

Permissible conclusion. Failure to recover a game means only that the tested procedure did not obtain a sufficiently predictive explanation under its stated evidence, assumptions, model class, and time budget.

The framework can nevertheless fail in informative ways. If removing communication does not affect the apparent cooperation, a strategic-communication hypothesis loses support. If conventional game models predict behavior accurately across held-out interventions, the need for more elaborate explanations weakens. If encoded games remain easy to identify, representational opacity may be less important than anticipated. If model-free containment consistently performs as well as theory-guided intervention, detailed provenance may have low immediate decision value in the tested settings. If the preferred risk heuristic produces greater loss than alternatives, it should be revised.

The strongest propositions should be stated in falsifiable form. Strategic identification is institutionally valuable only if it improves prediction, prevention, remediation, or redeployment decisions relative to credible baselines. Provenance is operationally material only if different supported explanations imply different actions or different expected consequences. Game theory is useful only to the extent that its explicit structure yields predictions and interventions that transfer beyond the observations used to construct the model.

Scientific credibility also requires that negative results remain visible. The project may find little spontaneous cooperation. It may find that most apparent coordination is common-cause correlation. It may find that established theory explains the tested behavior well. It may find that containment policies require little causal identification. Those results would not defeat the project. They would answer the questions more honestly than a forced dramatic conclusion.

Conclusion

Unexpected cooperation among artificial agents is important because institutions increasingly rely on systems whose behavior arises from interaction rather than from one isolated model response. The

relevant governance problem is not whether cooperation is inherently good or bad. It is whether the institution understands enough about the interaction to prevent unacceptable outcomes, respond before harm becomes irreversible, correct the mechanism that produced the incident, and decide responsibly whether the system may operate again.

Final proposition. The purpose of strategic diagnosis is not to eliminate uncertainty. It is to secure enough justified understanding to support prevention, safe response, effective remediation, and accountable authorization.

Game theory offers a disciplined first-principles language for this inquiry. It forces the analyst to specify players, actions, information, incentives, timing, beliefs, and equilibrium. Inverse game theory and related reward-learning research also warn that behavior may not uniquely reveal the objective or game that generated it. The appropriate goal is therefore not unlimited causal certainty. It is decision-relevant identification: enough justified understanding to improve an institutional choice, together with an honest account of residual uncertainty.

The paper does not presume that unexpected cooperation reflects unfamiliar intelligence, autonomous intent, or deliberate concealment. It does not claim that rationality establishes safety. It does not treat proposed simulations as completed evidence. It constructs a route through the problem: begin with ordinary explanations, distinguish correlation from strategic dependence, test predictions under intervention, compare theory-guided action with simple baselines, preserve provenance, and ask at every stage whether additional knowledge changes the decision.

The resulting risk-management principle is intentionally restrained. When time is scarce and consequences may become irreversible, containment should be guided by observed capability and effect rather than delayed by an unresolved theory of provenance. Once the immediate exposure is controlled, causal and strategic understanding can become essential to prevention, remediation, validation, and redeployment. Where identification remains impossible or too slow, uncertainty should be incorporated into permissions, monitoring, and the burden of proof for autonomy.

The ambition of this initial draft is not to close the discussion. It is to make the discussion precise enough that researchers can challenge its assumptions, institutions can recognize the decisions they face, and future simulations can produce conditional answers rather than theatrical demonstrations. The ultimate question is practical: what must a responsible institution know in order to act well when artificial agents cooperate in an unexpected way?

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